United States Patent Application Publication	20250287761
Kind Code	A1
Publication Date	September 11, 2025
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LIGHT-EMITTING DEVICE AND MANUFACUTRING METHOD OF LIGHT-EMITTING DEVICE

Abstract

A light-emitting device includes an insulating substrate having one or more first cutouts and one or more second cutouts provided on opposite side surfaces thereof. First and second wiring electrodes, which are separated from each other, correspond respectively to the first and second cutouts. The first and second wiring electrodes each cover a region that extends over an edge of the corresponding cutout on an upper surface of the substrate and a region that extends over an edge the corresponding cutout on a lower surface of the substrate from an internal surface of the cutout. First and second resin films cover surfaces the first and second wiring electrodes in regions that extend over edges along the first and second cutouts in a top view. A light-emitting element is placed on the substrate to allow power to be supplied by the first wiring electrode and the second wiring electrode.

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Appl. No.:	**18/858359**
Filed (or PCT Filed):	**April 10, 2023**
PCT No.:	**PCT/JP2023/014506**

Foreign Application Priority Data

JP	2022-070834	Apr. 22, 2022

Publication Classification

Int. Cl.: **H10H29/85** (20250101); **H10H29/01** (20250101)

U.S. Cl.:

CPC **H10H29/857** (20250101); **H10H29/0364** (20250101); **H10H29/8508** (20250101);

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a light-emitting device including a semiconductor light-emitting element and a manufacturing method of the same.

BACKGROUND ART

[0002] There has been known a surface-mounted LED called a chip LED as a small-sized light-emitting device in which a light-emitting element, such as a light emitting diode (LED), is used as a light source.

[0003] For example, Patent Document 1 discloses a surface-mounted light emitting diode as a light-emitting device having a structure in which a cathode electrode and an anode electrode formed on the upper surface of a glass epoxy substrate including a light-emitting diode element are routed to a back surface electrode formed on the back surface side of the glass epoxy substrate through respective through-hole electrodes disposed at both end portions of the glass epoxy substrate.

[0004] In addition, in the surface-mounted light emitting diode disclosed in Patent Document 1, it is described that masking tape is affixed to the upper surfaces of the through-hole electrodes, and a second resin is placed on the entire upper surface of the glass epoxy substrate including the top of the masking tape.

[0005] Patent Document 1: Japanese Unexamined Patent Application Publication No. 2000-261041

DISCLOSURE OF THE INVENTION

Problems to be Solved by the Invention

[0006] In the surface-mounted light emitting diode disclosed in Patent Document 1, the masking tape is used to stop the second resin from flowing into the through-hole electrodes during the formation of the second resin.

[0007] Similarly to the surface-mounted light emitting diode disclosed in Patent Document 1, if a second resin, which is a sealing resin, is formed on the entire upper surface of a glass epoxy substrate that has been subjected to thermosetting treatment by impregnating glass fibers with epoxy resin, the amount of resin material used may increase, and the cost may rise.

[0008] However, in a case where the structure is configured to suppress the amount of resin material used such that the resin material does not cover the through-hole electrodes in the light-emitting device, when individualization of the light-emitting device is performed by dicing, delamination may occur on the through-hole electrodes, the cathode electrode, the anode electrode, or the back surface electrode, which are exposed to dicing surfaces as side surfaces of the light-emitting device, causing a failure.

[0009] Furthermore, in a case where a plurality of pairs of electrodes are formed on the glass epoxy substrate, when the light-emitting device is mounted on a mounting substrate, solder may excessively crawl up through the through-hole electrodes. If the solder crawls onto the glass epoxy substrate, insufficient solder on the mounting surface may cause a failure in die shear strength.

[0010] The present invention has been made in consideration of the above-described points and an object of which is to provide a light-emitting device that allows improving mountability of the light-emitting device while suppressing a failure in manufacturing and a manufacturing method of the light-emitting device.

Solutions to the Problems

[0011] A light-emitting device according to the present invention includes an insulating substrate, a first wiring electrode, a second wiring electrode, a first resin film, a second resin film, and a light-emitting element. The insulating substrate has an approximately rectangular upper surface shape and having one or a plurality of first cutouts and one or a plurality of second cutouts. The one or the plurality of first cutouts are formed on one side surface of one set of opposed side surfaces. The one or the plurality of second cutouts are formed on another side surface of the one set of side surfaces. The first wiring electrode covers a region that extends over an edge of each of the one or the plurality of first cutouts on an upper surface of the substrate and a region that extends over an edge of each of the one or the plurality of first cutouts on a lower surface of the substrate from an internal surface of each of the one or the plurality of first cutouts. The second wiring electrode is separated from each of the first wiring electrodes. The second wiring electrode covers a region that extends over an edge of each of the one or the plurality of second cutouts on the upper surface of the substrate and a region that extends over an edge of each of the one or the plurality of second cutouts on the lower surface of the substrate from an internal surface of each of the one or the plurality of second cutouts. The first resin film covers a surface of each of the first wiring electrodes in each of regions that extend over edges along the one or the plurality of first cutouts in a top view viewed from a direction perpendicular to the upper surface of the substrate. The second resin film covers a surface of each of the second wiring electrodes in a region extending over edges along the one or the plurality of second cutouts in the top view. The light-emitting element is placed on the substrate to allow power to be supplied by the first wiring electrode and the second wiring electrode.

[0012] A manufacturing method of a light-emitting device according to the present invention includes: a step of preparing a substrate structure having an insulating substrate and copper foil layers formed on an upper surface and a lower surface of the substrate; a through hole forming step of forming a plurality of through holes arranged in a plurality of rows on the upper surface of the substrate structure, each of the plurality of through holes penetrating from the upper surface to the lower surface of the substrate structure; a first plating step of forming a copper layer on surfaces of the copper foil layers and inner surfaces of the plurality of through holes by plating; a wiring pattern forming step of forming a wiring pattern by etching each of the copper foil layers and the copper layer on the upper surface and the lower surface of the substrate; a resin film forming step of forming a resin film covering a region along an outer edge of each of the plurality of through holes on a surface of a part of the wiring pattern formed on the upper surface of the substrate structure; and an individualization step of individualizing by cutting the substrate structure so as to separate each of the plurality of through holes along dicing lines extending along the plurality of rows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. **1** is a top view of a light-emitting device according to an embodiment of the present invention.

[0014] FIG. **2** is a cross-sectional view of the light-emitting device according to the embodiment of the present invention.

[0015] FIG. **3** is a diagram illustrating a manufacturing process of the light-emitting device according to the embodiment of the present invention.

[0016] FIG. **4** is a diagram illustrating the manufacturing process of the light-emitting device according to the embodiment of the present invention.
[0017] FIG. **5**A is a top view of the light-emitting device according to the embodiment of the present invention in manufacturing.
[0018] FIG. **5**B is a cross-sectional view of the light-emitting device according to the embodiment of the present invention in manufacturing.
[0019] FIG. **6**A is a top view of the light-emitting device according to the embodiment of the present invention in manufacturing.
[0020] FIG. **6**B is a cross-sectional view of the light-emitting device according to the embodiment of the present invention in manufacturing.
[0021] FIG. **7** is a top view of the light-emitting device according to the embodiment of the present invention in manufacturing.
[0022] FIG. **8** is a cross-sectional view of the light-emitting device according to the embodiment of the present invention in manufacturing.
[0023] FIG. **9** is a top view of the light-emitting device according to the embodiment of the present invention in manufacturing.
[0024] FIG. **10** is a cross-sectional view illustrating an example of a mounted state of the light-emitting device according to the embodiment of the present invention.
[0025] FIG. **11**A is a top view of a light-emitting device according to a modification of the present invention.
[0026] FIG. **11**B is a cross-sectional view of the light-emitting device according to the modification of the present invention.

DESCRIPTION OF PREFERRED EMBODIMENTS

[0027] The following describes an embodiment of the present invention in detail. In the following description and attached drawings, same reference numerals are given to actually same or equivalent parts.

Embodiment

[0028] With reference to FIG. **1** and FIG. **2**, a configuration of a light-emitting device **100** according to an embodiment is described. FIG. **1** is a top view of the light-emitting device **100** according to the embodiment. FIG. **2** is a cross-sectional view taken along a line **2-2** of the light-emitting device **100** illustrated in FIG. **1**.
[0029] In FIG. **1**, a first resin film **31** and a second resin film **32** are indicated by two-dot chain lines in order to clarify the structure and positional relationship of each element on an upper surface of a substrate **10**. Additionally, in FIG. **1**, a sealing member **60** (see FIG. **2**) is not illustrated in order to clarify the structure and positional relationship of each element on the upper surface of the substrate **10**.

Light-Emitting Device

[0030] The light-emitting device **100** has the substrate **10** that has an approximately rectangular upper surface shape and has cutouts **13**A, **13**B, **13**C as three first cutouts and cutouts **15**A, **15**B, **15**C as three second cutouts formed on respective opposed sides. In the following description, the cutouts **13**A, **13**B, **13**C are collectively referred to as cutouts **13**A-C, and the cutouts **15**A, **15**B, **15**C are collectively referred to as cutouts **15**A-C in some cases.
[0031] In addition, the light-emitting device **100** has light-emitting elements **41**, **43**, **45** and protection elements **51**, **53** on the upper surface of the substrate **10**.
[0032] Moreover, the light-emitting device **100** has the sealing member **60** (see FIG. **2**) that seals the light-emitting elements **41**, **43**, **45** and the protection elements **51**, **53** on the upper surface of the substrate **10**.
[0033] In the embodiment, each component is placed in each of a plurality of light-emitting device formation regions arranged in a matrix on one flat plate substrate, and individualization by dicing is performed to manufacture a plurality of light-emitting devices **100**. Therefore, each side surface of

the light-emitting device **100** is a dicing surface cut by dicing.

[0034] The embodiment describes a case where the respective three light-emitting elements **41**, **43**, **45** are connected in parallel in the light-emitting device **100**. That is, the light-emitting device **100** of the embodiment is a light-emitting device that has three pairs of anode electrodes and cathode electrodes and can control the three light-emitting elements **41**, **43**, **45** individually.

Substrate

[0035] The substrate **10** is an insulating substrate having an approximately rectangular upper surface shape. The substrate **10** has a planar shape in which parts of one set of sides opposed to one another are cut out of a rectangle indicated by dashed lines in a top view. In other words, in the substrate **10**, cutouts are formed on parts of opposed side surfaces **10**S**1** and **10**S**2**. Hereinafter, the center line parallel to the above-described one set of sides of the substrate **10** is described as a center line CL.

[0036] The respective cutouts **13**A, **13**B, **13**C as the first cutouts and the respective cutouts **15**A, **15**B, **15**C as the second cutouts are formed at the center and both ends of each of the side surface **10**S**1** and the side surface **10**S**2** of the substrate **10** by penetrating from the upper surface to a lower surface of the substrate **10**.

[0037] The cutout **13**A, the cutout **13**B, and the cutout **13**C and the cutout **15**A, the cutout **15**B, and the cutout **15**C are formed at positions to be opposed across the center line CL.

[0038] The cutout **13**A and the cutout **15**A are formed at the respective centers of the side surfaces **10**S**1** and **10**S**2** in the top view in a shape obtained by cutting an approximately semicircular shape from the rectangular substrate **10**. In addition, the cutouts **13**B, **13**C and the cutouts **15**B, **15**C are formed at respective both ends of the side surfaces **10**S**1** and **10**S**2** in the top view in a shape obtained by cutting a fan shape from the rectangular substrate **10**.

[0039] In the embodiment, a flat plate made of glass epoxy was used as the substrate **10**. In addition, in the embodiment, a glass epoxy flat plate having a thickness of about 100 μm was used.

Wiring Electrode

[0040] Each of wiring electrodes **23**A, **23**B, **23**C as first wiring electrodes and wiring electrodes **25**A, **25**B, **25**C as second wiring electrodes is a conductive wiring electrode formed on a surface of the substrate **10**.

[0041] The respective wiring electrodes **23**A, **23**B, **23**C and the respective wiring electrodes **25**A, **25**B, **25**C are formed over respective edges of the corresponding cutouts **13**A-C and respective edges of the corresponding cutouts **15**A-C on the upper surface of the substrate **10**.

[0042] The wiring electrodes **23**A, **23**B, **23**C continuously extend from parts formed at the respective edges of the cutouts **13**A-C to bonding pad portions and element placing portions disposed in a central region along the center line CL of the substrate **10** via routing portions **23**AW, **23**BW, **23**CW, respectively. The wiring electrodes **25**A, **25**B, **25**C have parts formed at the respective edges of the cutouts **15**A-C, bonding pad portions disposed in the central region along the center line CL of the substrate **10**, and routing portions **25**AW, **25**BW, **25**CW that connect them. The wiring electrodes **23**A-C and the wiring electrodes **25**A-C are each separated from one another on the upper surface of the substrate **10**.

[0043] In addition, the wiring electrodes **23**A to **23**C and the wiring electrodes **25**A to **25**C are close to one another in respective regions along the center line CL between the cutouts **13**A to **13**C and the center line CL of the substrate **10** and between the cutouts **15**A to **15**C and the center line CL of the substrate **10** and extend from the regions along the center line CL to respective both end portions of the upper surface of the substrate **10**.

[0044] The wiring electrodes **23**A to **23**C are close to one another in a region along a straight line L**1** extending along an arranging direction of the cutouts **13**A to **13**C. Specifically, extension portions EX that extend along the straight line L**1** in respective directions toward the wiring electrodes **23**B and **23**C are formed at the routing portion **23**AW.

[0045] Further, the routing portions **23**BW and **23**CW of the wiring electrodes **23**B and **23**C are

formed to extend along the straight line L**1** in directions toward the wiring electrode **23**A from respective end portions of the upper surface of the substrate **10**. Thus, since the extension portions EX and the routing portions **23**AW to **23**CW of the wiring electrodes **23**A to **23**C have the parts extending along the straight line L**1**, the wiring electrodes **23**A to **23**C are configured closely in the region along the straight line L**1** on the upper surface of the substrate **10**.
[0046] The wiring electrodes **25**A to **25**C are close to one another in a region along a straight line L**2** extending along an arranging direction of the cutouts **15**A to **15**C. Specifically, an extension portion EX that extends along the straight line L**2** in a direction toward the wiring electrode **25**B is formed at the routing portion **25**AW. Further, an extension portion EX that extends along the straight line L**2** in a direction toward the wiring electrode **25**A is formed at the routing portion **25**CW.
[0047] In addition, the routing portion **25**BW of the wiring electrode **25**B is formed to extend along the straight line L**2** in a direction toward the wiring electrode **25**A from each end portion of the upper surface of the substrate **10**. Further, the routing portion **25**AW of the wiring electrode **25**A is formed such that a part opposed to the extension portion EX extends along the straight line L**2** toward the extension portion EX of the routing portion **25**CW. Thus, since the extension portions EX and the routing portions **25**AW to **25**CW of the wiring electrodes **25**A-C have the parts extending along the straight line L**2**, the wiring electrodes **25**A to **25**C are configured closely in the region along the straight line L**2** on the upper surface of the substrate **10**.
[0048] As described above, the wiring electrodes **23**A to **23**C and **25**A to **25**C are each configured to be close to one another at the extension portions EX, the routing portions **23**AW to **23**CW, and the routing portions **25**AW to **25**CW.
[0049] In addition, the respective wiring electrodes **23**A-C and the respective wiring electrodes **25**A-C reach a back surface of the substrate **10** from the respective edges of the cutouts **13**A-C and the cutouts **15**A-C on the upper surface of the substrate **10** through respective internal surfaces of the cutouts **13**A-C and the cutouts **15**A-C.
[0050] The wiring electrodes **23**A-C and the wiring electrodes **25**A-C are formed over the whole circumferences of the respective edges of the cutouts **13**A-C and the cutouts **15**A-C on the upper surface of the substrate **10**. In other words, the respective wiring electrodes **23**A-C and the respective wiring electrodes **25**A-C are formed to cover regions PA extending over the respective edges of the cutouts **13**A-C and the cutouts **15**A-C on the upper surface and the lower surface of the substrate **10**.
[0051] In the embodiment, each of the wiring electrodes **23**A-C and wiring electrodes **25**A-C is made of copper foil formed on substrate surfaces and copper plating formed on the surfaces of the copper foil on the upper surface and the lower surface of the substrate **10**. In addition, the wiring electrodes **23**A-C, **25**A-C are made of copper plating formed on the internal surfaces of the cutouts so as to cover the internal surfaces. In the embodiment, the thickness of each of the wiring electrodes **23**A-C and wiring electrodes **25**A-C on the upper surface and the lower surface of the substrate **10** is about 30 μm from the surface of the substrate **10**.
[0052] The shapes of the wiring electrodes **23**A-C and the wiring electrodes **25**A-C on the upper surface of the substrate **10** can be designed arbitrarily according to the arrangement of the bonding pad portions and the element placing portions in the central region along the center line CL of the substrate **10**. It is only necessary for the wiring electrodes **23**A-C and the wiring electrodes **25**A-C to be formed over the respective edges of the cutouts **13**A-C and the cutouts **15**A-C and extend up to respective both end portions of the upper surface of the substrate **10** while being close to one another at the regions along predetermined straight lines in the direction along the center line CL between the cutouts **13**A-C and the bonding pad portions with element placing portions and between the cutouts **15** A-C and the bonding pad portions with element placing portions in the central region.
Light-Emitting Element

[0053] The light-emitting elements **41**, **43**, **45** are light emitting diodes (LEDs) constituted of semiconductors placed on an upper surface of the wiring electrode **23**A in the central region of the upper surface of the substrate **10**.
[0054] In the embodiment, the light-emitting element **41** is an AlGaInP-based LED that emits red light from an upper surface. In the embodiment, the light-emitting elements **43** and **45** are InGaN-based LEDs that emit green and blue light from upper surfaces, respectively. The light-emitting elements **41**, **43**, **45** are bonded onto the upper surface of the wiring electrode **23**A via respective conductive element bonding layers (not illustrated) made from argentum (Ag) paste or the like as a raw material.
[0055] The light-emitting element **41** has an anode electrode on a lower surface and a cathode electrode pad on the upper surface. In the light-emitting element **41**, the anode electrode on the lower surface is electrically connected to the wiring electrode **23**A. Further, in the light-emitting element **41**, the cathode electrode pad on the upper surface is electrically connected to the wiring electrode **25**A by a bonding wire **41**K. Therefore, the light-emitting element **41** has a configuration that is supplied with electric power via the wiring electrode **23**A and the wiring electrode **25**A.
[0056] The light-emitting element **43** has a cathode electrode pad and an anode electrode pad on the upper surface. In the light-emitting element **43**, the cathode electrode pad is connected to the wiring electrode **23**B by a bonding wire **43**A, and the anode electrode pad is connected to the wiring electrode **25**B by a bonding wire **43**K. Therefore, the light-emitting element **43** has a configuration that is supplied with electric power via the wiring electrode **23**B and the wiring electrode **25**B.
[0057] Similarly to the light-emitting element **43**, the light-emitting element **45** has a cathode electrode pad and an anode electrode pad on the upper surface. In the light-emitting element **45**, the cathode electrode pad is connected to the wiring electrode **23**C by a bonding wire **45**A, and the anode electrode pad is connected to the wiring electrode **25**C by a bonding wire **45**K. Therefore, the light-emitting element **45** has a configuration that is supplied with electric power via the wiring electrode **23**C and the wiring electrode **25**C. The light-emitting element **43** and the light-emitting element **45** are bonded adjacent to the light-emitting element **41** on the upper surface of the wiring electrode **23**A via element bonding layers (not illustrated). A bonding surface between the light-emitting element **43** and the light-emitting element **45** is insulating and is electrically insulated from the wiring electrode **23**A.
Protection Element
[0058] The protection element **51** and the protection element **53** are reverse voltage protection elements, such as zener diodes. Each of the protection element **51** and the protection element **53** has a cathode electrode on a lower surface and an anode electrode pad on an upper surface.
[0059] In the protection element **51**, the cathode electrode on the lower surface is electrically connected to the wiring electrode **23**B. Further, in the protection element **51**, the anode electrode pad on the upper surface is electrically connected to the wiring electrode **25**B by a bonding wire **51**A.
[0060] In the protection element **53**, the cathode electrode on the lower surface is electrically connected to the wiring electrode **23**C. Further, in the protection element **53**, the anode electrode pad on the upper surface is electrically connected to the wiring electrode **25**C by the bonding wire **53**A.
[0061] Therefore, the protection element **51** and the protection element **53** are connected in parallel with the respective light-emitting elements **43**, **45** in reverse polarity between the respective wiring electrodes **23**B, **23**C and the respective wiring electrodes **25**B, **25**C.
First Resin Film, Second Resin Film, and Third Resin Film
[0062] The first resin film **31** (indicated by the two-dot chain line in the drawing) is an insulating film made of a resin material having an insulating property, such as epoxy resin. The first resin film **31** is formed to block each of the cutouts **13**A-C and cover the wiring electrodes **23**A, **23**B, **23**C at

peripheral edges of the cutouts **13**A, **13**B, **13**C on the upper surface of the substrate **10** in the top view.

[0063] The first resin film **31** further includes a strip-shaped portion **31**B that extends in a strip shape in a region along the straight line L**1**. The strip-shaped portion **31**B covers the upper surfaces of the extension portions EX and the routing portions **23**AW-CW of the wiring electrodes **23**A-C, which are formed in the region along the straight line L**1**, and the upper surface of the substrate **10**. That is, in the region along the straight line L**1**, the first resin film **31** covers the upper surface of each of the wiring electrodes **23**A-C with the strip-shaped portion **31**B and fills the gap between the respective wiring electrodes **23**A-C.

[0064] The second resin film **32** (indicated by the two-dot chain line in the drawing) is, similarly to the first resin film **31**, a film made of a resin material having an insulating property, such as epoxy resin. The second resin film **32** is formed to block each of the cutouts **15**A-C and cover the wiring electrodes **25**A, **25**B, **25**C at peripheral edges of the cutouts **15**A, **15**B, **15**C on the upper surface of the substrate **10** in the top view.

[0065] The second resin film **32** further includes a strip-shaped portion **32**B that extends in a strip shape in a region along the straight line L**2**. The strip-shaped portion **32**B covers the upper surfaces of the extension portions EX and the routing portions **25**AW-CW of the wiring electrodes **25**A-C, which are formed in the region along the straight line L**2**, and the upper surface of the substrate **10**. That is, in the region along the straight line L**2**, the second resin film **32** covers the upper surface of each of the wiring electrodes **25**A-C with the strip-shaped portion **32**B and fills the gap between the respective wiring electrodes **25**A-C.

[0066] In addition, on the upper surface of the substrate **10**, the first resin film **31** and the second resin film **32** covers the regions on the upper surfaces of the wiring electrodes **23**A-C and the wiring electrodes **25**A-C, respectively, corresponding to the regions PA of the substrate **10** covered by the wiring electrodes **23**A-C and by the respective wiring electrodes **25**A-C.

[0067] As illustrated in FIG. **2**, the first resin film **31** and the second resin film **32** enter the cutouts **13**A-C and the cutouts **15**A-C so as to fit into an inside of each of the cutouts **13**A-C and the cutouts **15**A-C from above, respectively. That is, the first resin film **31** and the second resin film **32** cover upper end parts of the wiring electrodes **23**A, **23**B, **23**C and the wiring electrodes **25**A, **25**B, **25**C in the cutouts **13**A, **13**B, **13**C and the cutouts **15**A, **15**B, **15**C, respectively.

[0068] As illustrated in FIG. **1**, in the light-emitting device **100**, the first resin film **31** and the second resin film **32** are formed to partially expose respective outer peripheral ends of the wiring electrodes **23**A-C and the wiring electrodes **25**A-C in the vicinities of the peripheral edges of the regions PA of the cutouts **13**A-C and the cutouts **15**A-C, respectively. This is because even when non-transparent solder resist ink is used for the first resin film **31** and the second resin film **32**, the positions of the wiring electrodes **23**A-C and the wiring electrodes **25**A-C and orientation of the light-emitting device **100** are viewable in the light-emitting device **100** after manufacturing.

[0069] As illustrated in FIG. **2**, a third resin film **33** is formed on the lower surface of the substrate **10**. The third resin film **33** is formed of the same material as that of the first resin film **31** and the second resin film **32**. In addition, the third resin film **33** is formed in a pattern that exposes bonding regions of the wiring electrodes **23**A-C and the wiring electrodes **25**A-C with bonding members for a mounting substrate on the lower surface of the substrate **10**.

[0070] In the embodiment, photosensitive solder resist ink (green resist) was used as the first resin film **31**, the second resin film **32**, and the third resin film **33**. The first resin film **31** and the second resin film **32** can avoid unintentional light by the light-emitting elements **41**, **43**, **45** by using a light-shielding member. This allows the light-emitting device **100** to obtain more desired light distribution. Note that the first resin film **31** and the second resin film **32** may be made of, for example, a member having light absorbency or light reflectivity, such as silicon. In the embodiment, each of the first resin film **31**, the second resin film **32**, and the third resin film **33** is formed such that the thickness from the upper surface of the substrate **10** is about 20 μm.

[0071] Nickel (Ni) and gold (Au) are stacked in sequence on the surface of each of the wiring electrodes **23**A-C and wiring electrodes **25**A-C exposed from the first resin film **31**, the second resin film **32**, and the third resin film **33**.

Sealing Member

[0072] The sealing member **60** is constituted of a translucent resin material, such as silicone resin, epoxy resin, and acrylic resin, and is formed in the region along the center line CL on the upper surface of the substrate **10**.

[0073] In the embodiment, the sealing member **60** is formed in the region between the vicinity of the straight line L**1** and the vicinity of the straight line L**2** on the upper surface of substrate **10**, covers the upper surface of the substrate **10** and the respective upper surfaces of the wiring electrodes **23**A-C, **25**A-C, and seals the light-emitting elements **41**, **43**, **45** and the protection elements **51**, **53**. That is, lower end portions of side surfaces in a direction along the center line CL of the sealing member **60** are on the respective upper surfaces of the strip-shaped portions **31**B and **32**B of the first resin film **31** and the second resin film **32** on the vicinities of the straight line L**1** and the straight line L**2** on the upper surface of the substrate **10**.

[0074] As described above, by setting the positions of the lower end portions of the side surfaces of the sealing member **60** to the respective upper surfaces of the strip-shaped portions **31**B and **32**B of the first and second resin films **31** and **32** and the vicinities on the straight line L**1** and the straight line L**2**, leakage of the sealing member **60** in directions of the respective cutouts can be suppressed in a manufacturing method described below.

Manufacturing Method of Light-Emitting Device

[0075] Next, using FIG. **3** to FIG. **9**, the manufacturing method of the light-emitting device **100** of the embodiment is described.

[0076] FIG. **3** and FIG. **4** are diagrams illustrating the manufacturing process of the light-emitting device **100** according to the embodiment of the present invention. In addition, FIG. **5** to FIG. **9** illustrate cross-sectional views of the light-emitting device **100** at each step of the manufacturing procedure illustrated in FIG. **3** and FIG. **4**. FIG. **5**A, FIG. **6**A, FIG. **7**, and FIG. **9** illustrate a top view of one light-emitting device formation region and its periphery. FIG. **5**B, FIG. **6**B, and FIG. **8** illustrate a cross-sectional surface corresponding to the line **2-2** illustrated in FIG. **1** at the one light-emitting device formation region and its periphery.

[0077] In the description of the manufacturing method below, the cutouts **13**A-C, **15**A-C of the substrate **10** before the completion of Step S**204** (individualization step) are referred to as through holes **13**A-C, **15**A-C, respectively.

Step S**101**

[0078] First, a step of preparing a substrate structure **10**M (see FIGS. **5**A and **5**B) used for manufacturing is performed (Step S**101**, substrate structure preparation step). The substrate structure **10**M is a substrate formed by press-bonding a copper foil CU**1** onto the upper surface of the substrate **10**, which is a flat plate made of glass epoxy having a rectangular upper surface shape, and a copper foil CU**2** onto the lower surface. Dicing lines DL**1**, DL**2** are dicing lines used to individualize the light-emitting device **100**. That is, the light-emitting device **100** is formed by individualizing by dicing a structure including the substrate structure **10**M along the dicing lines DL**1**, DL**2**. The copper foils CU**1**, CU**2**, for example, are press-bonded onto a glass epoxy substrate and have a thickness of about 18 μm.

Step S**102**

[0079] Next, as illustrated in FIGS. **5**A and **5**B, in each light-emitting device formation region, through holes, which become the cutouts **13**A-C and the cutouts **15**A-C, are formed along the dicing lines DL**1** (Step S**102**, cutout forming step). In this step, the through holes are formed along each of the dicing lines DL**1**. The through holes **13**B, **13**C, **15**B, and **15**C are formed at positions where the dicing lines DL**1** intersect with the dicing lines DL**2**, and the through holes **13**A and **15**A are formed at the middle points of the adjacent dicing lines DL**2**. Each of the through holes **13**A-C and **15**A-C was formed by penetrating the copper foil CU**1**, the substrate structure **10**M, and the copper foil CU**2** from above by means of a drill or the like.

Step S**103**

[0080] Next, as illustrated in FIGS. **6**A and **6**B, copper plating is applied to inner surfaces of the through holes of the substrate structure **10**M and surfaces of the copper foil CU**1** and the copper foil CU**2** to form a copper layer CU**3** that integrally covers the surface of the substrate structure **10**M (Step S**103**, first plating step). In this step, first, copper electroless plating was applied to the substrate structure **10**M to form electroless plating layers on the surfaces of the copper foil CU**1** and the copper foil CU**2** and the inner surfaces of the through holes of the substrate structure **10**M. Next, copper electrolytic plating was applied to this substrate structure **10**M to form an electrolytic plating layer on the surface of the electroless plating layers, and the copper layer CU**3** was formed. The copper layer CU**3** includes the copper foil CU**1** and the copper foil CU**2** as foundation layers on the upper surface and the lower surface of the substrate **10**.

Step S**104**

[0081] Next, as illustrated in FIG. **7**, the copper layer CU**3** is etched to form the wiring electrodes **23**A-C and the wiring electrodes **25**A-C in each light-emitting device formation region (Step S**104**, wiring pattern forming step). In this step, first, a resist mask was formed on the surface of the copper layer CU**3** of the substrate structure **10**M so as to be the wiring pattern described above. Then, the copper layer CU**3** exposed from the resist mask was etched until the surface of the substrate structure **10**M was exposed, and each of the wiring electrodes **23**A-C, **25**A-C was formed. Afterwards, the resist mask on the surface of each wiring electrode was removed.

[0082] At this time, the wiring electrodes **23**A-C and the wiring electrodes **25**A-C opposed between adjacent light-emitting device formation regions were formed to be continuous.

Step S**105**

[0083] Next, as illustrated in FIG. **8**, photosensitive resin RL is applied to the entire upper surface of the substrate structure **10**M (Step S**105**, resin film application step). In this step, the resin RL was applied to the upper surface of the substrate structure **10**M collectively by squeegeeing from above the substrate structure **10**M using a screen printing machine. At this time, the viscosity, application amount, or squeegeeing condition of the resin RL are set such that the resin RL is applied to (the resin RL enters) the respective upper end parts of the wiring electrodes **23**A, **23**B, **23**C and the wiring electrodes **25**A, **25**B, **25**C in the respective cutouts **13**A, **13**B, **13**C and the respective cutouts **15**A, **15**B, **15**C.

Step S**106**

[0084] Next, as illustrated in FIG. **9**, the resin RL applied to the substrate structure **10**M is exposed to light to form the first resin film **31** and the second resin film **32** (Step S**106**, resin film pattern forming step). In this step, ultraviolet rays were irradiated from above the substrate structure **10**M through a photomask using an exposure device to expose the resin RL to ultraviolet rays in the shapes of the first resin film **31** and the second resin film **32** in each light-emitting device formation region.

[0085] At this time, the photomask has a pattern in which the resin RL is irradiated with ultraviolet rays in regions of the resin RL that cover the regions of the cutouts **13**A-C and the cutouts **15**A-C and the regions PA of the substrate **10**.

[0086] Furthermore, the photomask has a pattern in which the resin RL is irradiated with ultraviolet rays such that, for each row of a plurality of light-emitting device formation regions arranged in the direction along the dicing lines DL**1** on which the respective cutouts are formed, the strip-shaped portions **31**B and **32**B of the first resin film **31** and the second resin film **32** are continuous in regions along the straight lines L**1** and L**2** up to light-emitting device formation regions at both end portions of the row.

[0087] Afterwards, the resin RL in an unexposed portion was removed to form the first resin film **31** and the second resin film **32**.

[0088] In the embodiment, the case where it is controlled to apply the resin RL to the respective upper end parts of the wiring electrodes **23**A-C, **25**A-C in the cutouts **13**A-C, **15**A-C in Step S**105** (resin film application step) has been described.
[0089] In Step S**106** (resin film pattern forming step), the depth of focus of the exposure device may be adjusted to expose the resin RL to light to a desired depth range.
[0090] In addition, the third resin film **33** was formed on the lower surface of the substrate structure **10**M following Step S**106** (resin film pattern forming step). For the formation of the third resin film **33**, the resin RL was applied using a metal mask or the like that was opened in the shape of the third resin film **33** during squeegeeing.
Step S**107**
[0091] Next, plated layers, in which Ni and Au are stacked in sequence on the surfaces of the wiring electrodes **23**A-C and the wiring electrodes **25**A-C exposed from the respective resin films of the substrate structure **10**M, are formed (Step S**107**, second plating process). In this step, after Ni layers were formed on the surfaces of the wiring electrodes **23**A-C and the wiring electrodes **25**A-C by electrolytic plating, Au layers were formed on the surfaces of the Ni layers.
Step S**201**
[0092] Next, as illustrated in FIG. **1**, on the upper surface of each light-emitting device formation region, the light-emitting elements **41**, **43**, **45** and the protection elements **51**, **53** are bonded to predetermined positions (Step S**201**, element bonding step). In this step, first, the substrate structure **10**M was set on a die bonding apparatus, and raw material paste of a conductive element bonding layer was applied to each element placing portion on the upper surface of each of the wiring electrodes **23**A-C. Next, each element was placed on the raw material paste applied to each element placing portion with each electrode pad facing upward. Afterwards, the substrate structure **10**M was heated to harden the raw material paste, and each element was bonded to each wiring electrode **23**A-C.
Step S**202**
[0093] Next, as illustrated in FIG. **1**, in each light-emitting device formation region, bonding wires that connect the electrode pad of each element to each wiring electrode are formed (Step S**202**, wire bonding step). In this step, the substrate structure **10**M was set on a bonding device to form each of the bonding wires.
[0094] In the light-emitting element **41**, first, a metal bump was formed on the electrode pad of the light-emitting element **41** using a metal wire of Au or the like, and then the metal wire was bonded to a bonding pad portion and a metal bump of the wiring electrode **25**A to form the bonding wire **41**K. Subsequently, the bonding wires **43**A, **43**K, **45**A, **45**K, **51**A, and **53**A were formed using the same method.
Step S**203**
[0095] Next, as illustrated in FIG. **2**, the sealing member **60**, which seals each of the light-emitting elements and protection elements, is formed (Step S**203**, sealing member forming step). In this step, the substrate structure **10**M was set in a mold of a molding device, and raw material resin of the sealing member **60** was injected and hardened to form the sealing member **60**.
[0096] In the embodiment, for each row of a plurality of light-emitting device formation regions arranged in the direction along the dicing lines DL**1** on which the respective cutouts were formed, a mold with a recessed portion was used. The recessed portion was formed in the shape of the sealing member **60** continuously up to light-emitting device formation regions at both end portions of the row. That is, the end portions of the recessed portion of the mold on the upper surface side of the substrate structure **10**M are positioned in the vicinities of the straight lines L**1** and L**2** extending over the plurality of light-emitting device formation regions arranged in the above-described row direction.
[0097] Accordingly, when the substrate structure **10**M is pressed by the mold, the elasticity of the first and second resin films **31**, **32** allows the mold and the first and second resin films **31**, **32** to be

closely contacted even in gap portions between the respective wiring electrodes in the vicinities of the straight lines L**1** and L**2**. Therefore, when the raw material resin of the sealing member **60** is injected, leakage of the raw material resin of the sealing member **60** from the gap portions can be suppressed.

Step S**204**

[0098] Next, as illustrated in FIG. **1**, dicing is performed along the dicing lines DL**1** and DL**2** to individualize a plurality of light-emitting devices **100** (Step S**204**, individualization step). In this step, the substrate structure **10**M, which was affixed to a dicing sheet, was set to a dicing machine, and a dicing blade was moved in directions of predetermined traveling directions along the dicing lines DL**1** and DL**2** while being rotated to cut off the substrate structure **10**M. That is, dicing was performed to separate the cutouts **13**A-C and the cutouts **15**A-C formed on the dicing lines DL**1** of each substrate structure **10**M.

[0099] In the manufacturing method of the light-emitting device **100** as described above, the first resin film **31** and the second resin film **32** are formed to cover the upper surfaces of the wiring electrodes **23**A-C, **25**A-C on the dicing lines DL**1** and DL**2** and fit into the inside of each of the cutouts **13**A-C and the inside of each of the cutouts **15**A-C from above.

[0100] In other words, the first resin film **31** and the second resin film **32** are formed to cover the regions corresponding to the regions PA of the substrate **10** on the upper surfaces of the wiring electrodes **23**A-C, **25**A-C, especially the edges of the regions PA along the side surfaces **10**S**1** and **10**S**2**, respectively, after the individualization of the light-emitting device **100**. That is, the first resin film **31** and the second resin film **32** are holding members of the wiring electrodes **23**A-C, **25**A-C.

[0101] This can suppress delamination of each wiring electrode from the substrate **10** by the dicing blade during dicing when the light-emitting devices **100** are individualized. As described above, the first resin film **31** and the second resin film **32** are formed such that they fit into the cutouts in the finished light-emitting device **100** and into the through holes during manufacturing. Accordingly, delamination of the first resin film **31** and the second resin film **32** themselves can be avoided during the individualization, and the loss of delamination prevention effect of the wiring electrodes due to the delamination of the resin films can be avoided. This can suppress a failure in manufacturing the light-emitting device **100**.

Mounted State of Light-Emitting Device

[0102] FIG. **10** is a cross-sectional view of the light-emitting device **100** according to the embodiment in a mounted state. Note that FIG. **10** is a cross-sectional view taken along the line **2-2** of the light-emitting device **100** illustrated in FIG. **1**.

[0103] A mounting substrate **200** is a substrate on which the light-emitting device **100** is mounted. The mounting substrate **200** has a mounting electrode **201** and a mounting electrode **202** on the cross-sectional surface along the line **2-2**. Bonding members **203**, **204** are bonding members made from paste containing solder and the like as a raw material.

[0104] The bonding members **203**, **204** are heated and melted to bond the mounting substrate **200** and the light-emitting device **100** on upper surfaces of the mounting electrode **201** and the mounting electrode **202** of the mounting substrate **200**.

[0105] At the time of melting, the bonding members **203**, **204** spread to and wet the respective surfaces of the wiring electrode **23**A and the wiring electrode **25**A of the light-emitting device **100**. That is, the respective bonding members **203** and **204** crawl up along the inner surfaces of the wiring electrode **23**A and the wiring electrode **25**A formed on the inner surfaces of the cutout **13**A and the cutout **15**A.

[0106] In the light-emitting device **100**, the first resin film **31** and the second resin film **32** made of solder resist, which is a material with low affinity with a metal constituting the melted bonding members **203**, **204**, block the cutouts **13**A-C and the cutouts **15**A-C of the substrate **10**, respectively. Accordingly, with the light-emitting device **100** and the manufacturing method thereof, when the light-emitting device **100** is mounted on a mounting substrate **200**, the melted bonding members **203** and **204** crawling up to the upper surface through the internal surfaces of the cutouts can be avoided, and a resulting failure, such as a short circuit of wiring, can be avoided.

[0107] In addition, by avoiding the bonding members **203** and **204** excessively crawling up by the first resin film **31** and the second resin film **32**, thick fillets of the bonding members **203** and **204** can be formed.

[0108] The first and second resin films **31**, **32** preferably enter the insides of the cutouts up to a distance D, which is ⅓ or more and ½ or less of a thickness T from upper ends to lower ends of the wiring electrodes **23**A-C, **25**A-C ($T/3 \leq D \leq T/2$).

[0109] This is because if the distance D is too large, that is, if the resin films excessively enter the insides of the cutouts, the area of each wiring electrode exposed to the inner surface of each of the cutouts **13**A-C and cutouts **15**A-C decreases, which may cause the bonding strength with a bonding member (solder) to decrease when the light-emitting device **100** is mounted.

[0110] Conversely, if the distance D is too small, that is, if entry of the resin films into the cutouts is too small, the first and second resin films **31**, **32** are more likely to delaminate, which may cause the above-described delamination prevention effect of wiring and crawling up prevention effect of the bonding member to be lost.

[0111] As described above, with the light-emitting device **100** of the embodiment, the mounted light-emitting device **100** can secure high die shear strength and improve mountability.

Modification

[0112] Next, a light-emitting device **100**A of a modification is described. FIG. **11**A is a top view of the light-emitting device **100**A according to the modification. FIG. **11**B illustrates a cross-sectional surface of the light-emitting device **100**A corresponding to the line **2-2** illustrated in FIG. **1**.

[0113] The light-emitting device **100**A of the modification basically has a similar configuration to the light-emitting device **100** of the embodiment.

[0114] The light-emitting device **100**A of the modification differs from the embodiment in that a first resin film **71** and a second resin film **72** do not block the respective cutouts **13**A-C, **15**A-C in a top view.

[0115] For example, the light-emitting device **100**A of the modification can be achieved by forming through holes that penetrate through respective predetermined regions of the through holes **13**A-C and **15**A-C of the substrate structure **10**M from above by means of a drill or the like after Step S**203** (sealing member forming step) illustrated in FIG. **4**.

[0116] Since the light-emitting device **100**A of the modification has a structure in which the first and second resin films **71**, **72** are not provided at four corners of an individual light-emitting device formation region of the substrate structure **10**M in the top view, chipping of the first and second resin films **71**, **72** can be suppressed in Step S**204** (individualization step).

[0117] Even when the first resin film **71** and the second resin film **72** are formed as described above, the effects similar to those of the embodiment can be provided.

[0118] Therefore, with a light-emitting device **100**B of the modification, it is possible to improve mountability when the light-emitting device **100**B is mounted while suppressing failures in manufacturing the light-emitting device **100**B.

[0119] Note that in the embodiment and the modification, the cutouts **13**A-C and the cutouts **15**A-C each have semicircular and fan shapes, but the shapes of the respective cutouts are not limited thereto. The cutouts **13**A-C and the cutouts **15**A-C may each be formed in semi-oval and quarter-oval shapes or in semi-elliptical and quarter-elliptical shapes.

[0120] In the embodiment and the modification, while the first resin film **31** and the second resin film **32** are solder resist ink, for example, transparent or translucent resin films may be used. In this case, the first resin film **31** and the second resin film **32** may be formed in the entire region on the upper surface of the substrate **10**, excluding the element placing portions and the bonding portions of each of the wiring electrodes **23**A-C and wiring electrodes **25**A-C.

[0121] Thus, the embodiments described are not intended to limit the scope of the invention. The embodiments described can be performed in various other configurations, and various kinds of omission, replacement, and changes can be made without departing from the gist of the invention. Then, the modifications thereof are also included in the scope and gist of the invention, and are included in the scope of the invention and the equivalents described in the claims.

DESCRIPTION OF REFERENCE SIGNS

[0122] **100** Light-emitting device [0123] **10** Substrate [0124] **13**, **15** Cutout [0125] **23**, **25** Wiring electrode [0126] **31**, **71**, **81** First resin film [0127] **32**, **72**, **82** Second resin film [0128] **41**, **43**, **45** Light-emitting element [0129] **51**, **53** Protection element [0130] **60** Sealing member [0131] **200** Mounting substrate [0132] **201**, **202** Mounting electrode [0133] **203**, **204** Bonding member

Claims

1. A light-emitting device comprising: an insulating substrate having an approximately rectangular upper surface shape and having a plurality of first cutouts and a plurality of second cutouts, the plurality of first cutouts being formed on one side surface of one set of opposed side surfaces of the substrate, and the plurality of second cutouts being formed on another side surface of the one set of side surfaces of the substrate; a plurality of first wiring electrodes corresponding respectively to the plurality of first cutouts and each covering a region that extends over an edge of a corresponding one of the plurality of first cutouts on an upper surface of the substrate and a region that extends over an edge of the corresponding one of the plurality of first cutouts on a lower surface of the substrate from an internal surface of the corresponding one of the plurality of first cutouts; a plurality of second wiring electrodes separated from each of the first wiring electrodes and corresponding respectively to the plurality of second cutouts, each of the second wiring electrodes covering a region that extends over an edge of a corresponding one of the plurality of second cutouts on the upper surface of the substrate and a region that extends over an edge of the corresponding one of the plurality of second cutouts on the lower surface of the substrate from an internal surface of the corresponding one of the plurality of second cutouts; a first resin film covering a surface of each of the first wiring electrodes in each of regions that extend over edges along the plurality of first cutouts in a top view viewed from a direction perpendicular to the upper surface of the substrate; a second resin film covering a surface of each of the second wiring electrodes in a region extending over edges along the plurality of second cutouts in the top view; and a light-emitting element placed on the substrate to allow power to be supplied by the first wiring electrode and the second wiring electrode.

2. The light-emitting device according to claim 1, wherein: the first resin film extends up to a surface of a part of each of the first wiring electrodes formed on the internal surface of each of the plurality of first cutouts, and the second resin film extends up to a surface of a part of each of the second wiring electrodes formed on the internal surface of each of the plurality of second cutouts.

3. The light-emitting device according to claim 1, wherein the first resin film and the second resin film are formed to block the plurality of first cutouts and the plurality of second cutouts, respectively, in the top view.

4. The light-emitting device according to claim 2, wherein: the first resin film coats from an upper end of a part of each of the first wiring electrodes formed on the internal surface of each of the plurality of first cutouts up to a region at a distance of ⅓ or more and ½ or less of a thickness from an upper end to a lower end of the first wiring electrode, and the second resin film coats from an upper end of a part of each of the second wiring electrodes formed on the internal surface of each of the plurality of second cutouts up to a region at a distance of ⅓ or more and ½ or less of a thickness from an upper end to a lower end of the second wiring electrode.

5. The light-emitting device according to claim 1, wherein: each of the first wiring electrodes and the second wiring electrodes is made of copper, and the first resin film and the second resin film are formed to be in contact with the copper of the first wiring electrodes and the second wiring electrodes.

6. The light-emitting device according to claim 5, wherein nickel and gold are stacked in sequence on respective surfaces of the first wiring electrodes and the second wiring electrodes exposed from the first resin film and the second resin film.

7. The light-emitting device according to claim 1, wherein the plurality of first cutouts with the first wiring electrodes and the plurality of second cutouts with the second wiring electrodes are formed to be paired up with one another on the one set of side surfaces of the substrate, and a plurality of the light-emitting elements are placed on the substrate according to a number of the pairs.

8. The light-emitting device according to claim 1, further comprising: a sealing member formed to seal the light-emitting element on the upper surface of the substrate, the sealing member being formed to expose each of regions on respective surfaces of the first wiring electrodes extending over edges along the plurality of first cutouts and on respective surfaces of the second wiring electrodes extending over edges along the plurality of second cutouts.

9. The light-emitting device according to claim 8, wherein: the first resin film has a strip-shaped portion extending in a strip shape across both end portions of the substrate in a direction along the one set of side surfaces of the substrate between the plurality of first cutouts and the light-emitting element, the second resin film has a strip-shaped portion extending in a strip shape across both end portions of the substrate in a direction along the one set of side surfaces of the substrate between the plurality of second cutouts and the light-emitting element, each of the first wiring electrodes has a part extending along the strip-shaped portion of the first resin film at a lower portion of the first resin film, each of the second wiring electrodes has a part extending along the strip-shaped portion of the second resin film at a lower portion of the second resin film, and the sealing member is formed such that one of opposed side end portions thereof is placed on the strip-shaped portion of the first resin film and another of the opposed side end portions is placed on the strip-shaped portion of the second resin film.

10. A manufacturing method of the light-emitting device according to claim 1, comprising: preparing a substrate structure having an insulating substrate and copper foil layers formed on an upper surface and a lower surface of the substrate; forming a plurality of through holes arranged in a plurality of rows on the upper surface of the substrate structure, each of the plurality of through holes penetrating from the upper surface to the lower surface of the substrate structure; forming a copper layer on surfaces of the copper foil layers and inner surfaces of the plurality of through holes by plating; forming a wiring pattern by etching each of the copper foil layers and the copper layer on the upper surface and the lower surface of the substrate; forming a resin film covering a region along an outer edge of each of the plurality of through holes on a surface of a part of the wiring pattern formed on the upper surface of the substrate structure; and cutting the substrate structure so as to separate each of the plurality of through holes along dicing lines extending along the plurality of rows.

11. The manufacturing method of a light-emitting device according to claim 10, wherein forming the resin layer comprises: applying photosensitive resin to the upper surface of the substrate and regions along upper ends of internal surfaces in the wiring pattern formed on inner surfaces of the plurality of through holes by squeegeeing; and exposing the resin to light to form a resin film covering up to regions along outer edges of the plurality of through holes on a surface of the part of the wiring pattern formed on the upper surface of the substrate structure and surfaces of parts of the wiring pattern formed on the inner surfaces of the plurality of through holes.